

Some Differences Between Fields

- In biology, theories about some emergent phenomena (e.g., Darwin's theory of species) came first.
 - No accepted 'equations of motion' for constituent parts. A few explicit 'microphysics' calculations been attempted (evolutionary dynamics) with synthetic equations.
 - There are natural hierarchies — simpler systems (drosophila) — to help understand.
- In statistical physics, the scale separation between macrophysical phenomena (e.g., temperature) and microphysics (e.g., bouncing molecules) is very large.
 - Sensible to have theories of both: thermodynamics and kinetic theory.
- In astrophysics
 - Similar to geosciences in some ways. Well accepted equations of motion, complicated phenomena.
 - Sometimes use numerics just to flesh out a theory and make it testable (e.g., general relativity). No parameterizations in that case.
- In geosciences and climate, large computers allow direct computation of complex phenomena (e.g., the climate system, the Gulf Stream).
 - Well accepted equations of motion.
 - No natural hierarchies — just one climate system. So we must construct our own synthetic hierarchies.